Codefair.io: An Open Source and Free Software for Easily Making GitHub Repositories FAIR

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Making Software FAIR

Making Software FAIR

FAIR4RS principles

FAIR Principles for Research Software or FAIR4RS Principles (2022)
provide instructions for making software findable and reusable

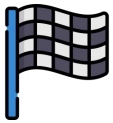
F1. Software is assigned a globally unique and persistent identifier.	How do I assign a unique identifier?
F1.1. Components of the software representing levels of granularity are assigned distinct identifiers.	
F1.2. Different versions of the software are assigned distinct identifiers.	
F2. Software is described with rich metadata.	What is rich metadata?
F3. Metadata clearly and explicitly include the identifier of the software they describe.	
F4. Metadata are FAIR, searchable and indexable.	

(and more)

https://doi.org/10.1038/s41597-022-01710-x

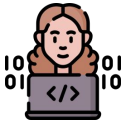
Making Software FAIR

FAIR-BioRS (2023): Actionable guidelines



Before starting

- Work from a version controlled system platform (e.g. GitHub)
- Select a license and include a LICENSE file



While developing the software

- Record dependencies (e.g. requirements.txt for Python)
- Maintain a README

When releasing a new version

- Include metadata in codemeta.json, CITATION.cff, CHANGELOG
- Archive source code on a DOI-issuing repository like Zenodo

Full guidelines: https://doi.org/10.1038/s41597-023-02463-x

Making Software FAIR
Challenges

FAIR-BioRS guidelines

Lack inputs outside of biomedical to be generalized → Actionable FAIR4RS Task Force

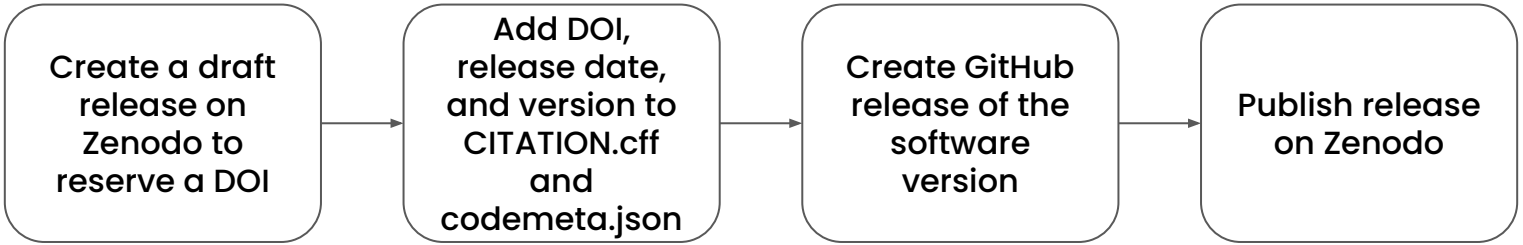
www.researchsoft.org/tf-actionable-fair4rs

Time consuming and difficult to implement →

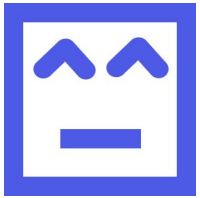
Making Software FAIR

Example of a challenge

Making and keeping software FAIR requires to
follow multiple steps for each version release of a software



Codefair



Codefair
Overview

Open source and free platform that acts as your
personal assistant for making software FAIR at the source

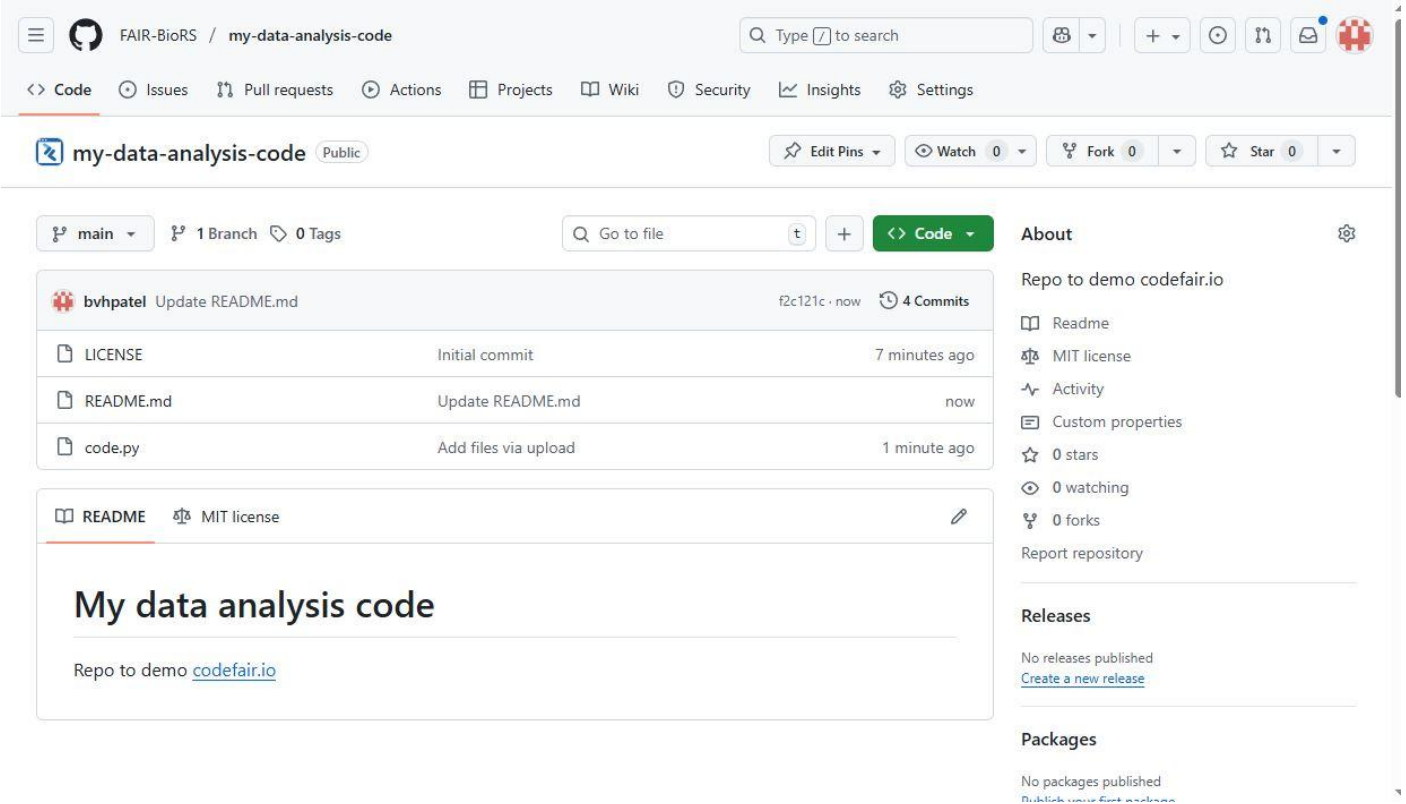
Started in 2024 as a side project mostly to help our team

Received $20k in Azure credits from the Microsoft AI for Good Lab

We chose GitHub as our first integration point

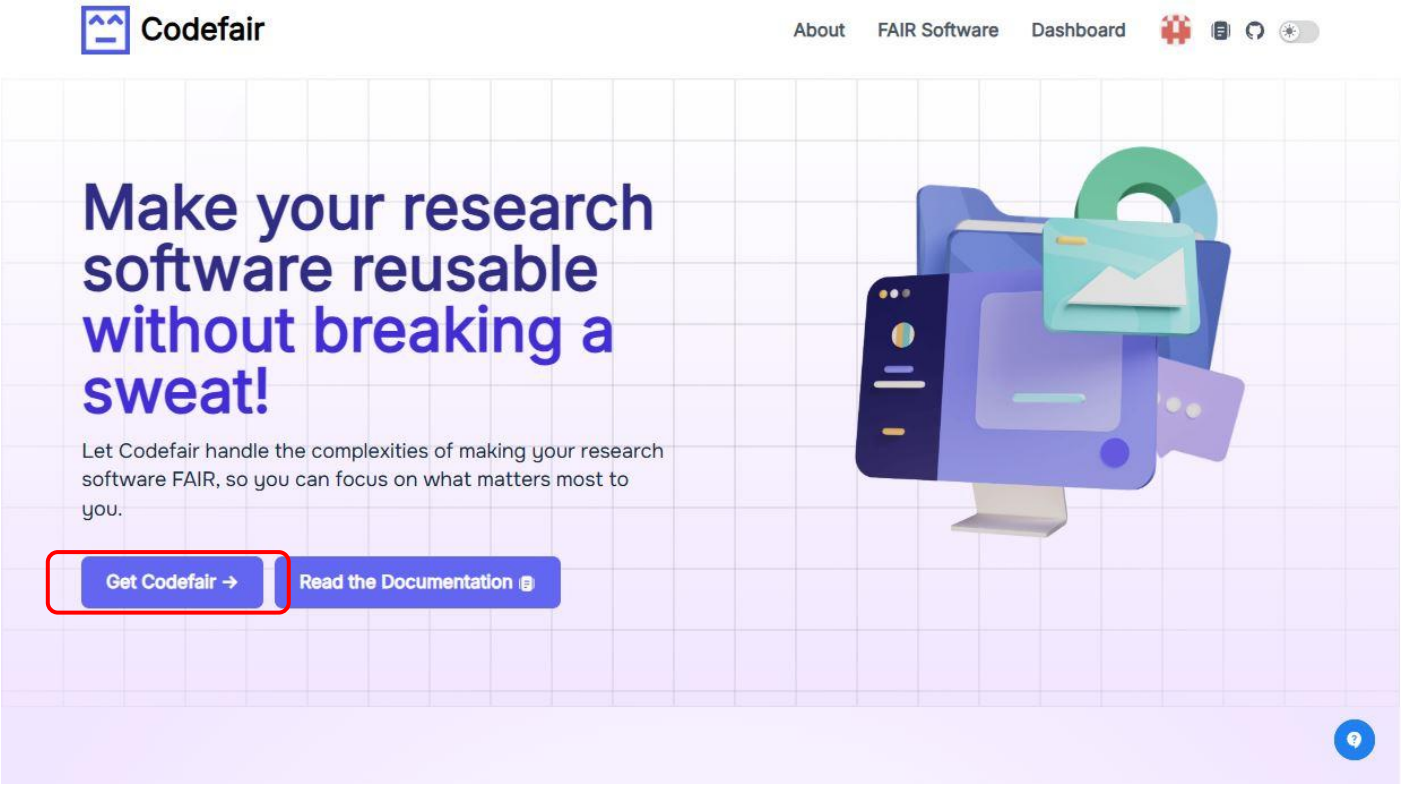
Codefair

Demo



Codefair

Demo



Codefair

Demo

Marketplace / Apps

Type / to search

codefair.io App

Add

README Transparency

Codefair is your personal assistant for making research software reusable and compliant with the Findable, Accessible, Interoperable, Reusable (FAIR) Principles for Research Software. Whether you're developing AI/ML models with Python, data visualization tools with Jupyter, or data analysis code with R, Codefair is here to help.

Make your research software reusable without breaking a sweat!

By combining a GitHub issue dashboard, a website with intuitive user interfaces, and automated pull request, Codefair ensures your software follows best practices, provides standard metadata, includes a license file, is archived on Zenodo, and more. All you need to do is:

1. Install codefair-app on the GitHub organizations/repositories of your choice (click "Install it for free" below)
2. Code as usual
3. Track FAIR compliance issues through the issue dashboard and address them through the Codefair website

With Codefair by your side, you're not just developing software but you're advocating for better software practices.

Note

While Codefair is free, installing it via the GitHub Marketplace may still require a credit card to be associated with your (or your organization's) GitHub account. To circumvent this, you can alternatively install Codefair directly from the app page.

About

Codefair is your personal assistant when it comes to making your research software reusable

By fairdataihub 40 installs

Verified

GitHub has verified the publisher's identity, ownership of their domain, and compliance with other requirements.

Tags 2

code-quality utilities

Pricing

Basic plan available.

Resources

Support

Documentation

Privacy Policy

Terms of Service

Report abuse

Codefair
Demo

Archive
Logs
Deleted repositories
Developer settings

Read access to metadata
Read and write access to code, issues, and pull requests

Repository access

All repositories
This applies to all current and future repositories owned by the resource owner. Also includes public repositories (read-only).

Only select repositories
Select at least one repository. Also includes public repositories (read-only).

Select repositories

Selected 2 repositories.

FAIR-BioRS/my-data-analysis-code
FAIR-BioRS/Hub

Save Cancel

Danger zone

Suspend your installation
This will block the app access to your resources.
Suspend

Uninstall "codefair.io"

Codefair

Demo

FAIR-BioRS / my-data-analysis-code

Type / to search

Code | Issues 1 | Pull requests | Actions | Projects | Wiki | Security | Insights | Settings

FAIR Compliance Dashboard #1

Edit | New issue

Open

codefair-io bot opened now

Check the FAIRness of your software

This issue is your repository's dashboard for all things FAIR. Keep it open as making and keeping software FAIR is a continuous process that evolves along with the software. You can read the documentation to learn more.

README ✔

A `README.md` file was found at within your repository.

Edit README

LICENSE ✔

A `LICENSE` file is found at the root level of the repository.

Edit License

Metadata ✘

To make your software FAIR, a `CITATION.cff` and `codemeta.json` are expected at the root level of your repository. These files are not found in the repository. If you would like Codefair to add these files, click the "Add metadata" button below to go to our interface for providing metadata and generating these files.

Add Metadata

FAIR Software Release ✘

To make your software FAIR, it is necessary to archive it in an archival repository like Zenodo every time you make a release. When you are ready to make your next release, click the "Create release" button below to easily create a FAIR release where your metadata files are updated (including with a DOI) before creating a GitHub release and archiving it.

Create Release

Assignees
No one - Assign yourself

Labels
No labels

Type
No type

Projects
No projects

Milestone
No milestone

Relationships
None yet

Development
Code with agent mode
Create a branch for this issue or link a pull request.

Notifications | Customize
Subscribe
You're not receiving notifications from this thread.

Participants
No participants

Transfer issue
Duplicate issue

Codefair
Demo

Edit metadata for my-data-analysis-code

Need help?

To make your software FAIR, a CITATION.cff and codemeta.json file are expected at the root level of your repository. They help people discover your software and provide information about your software to them. Provide metadata about your software below and codefair will submit a pull request with a CITATION.cff and codemeta.json file for you.

Basic Information

General information of the repository.

Software Name *

my-data-analysis-code

Description *

Repo to demo codefair.io

Creation Date

2025-09-30

First Release Date

Select Date

Authors and Contributors

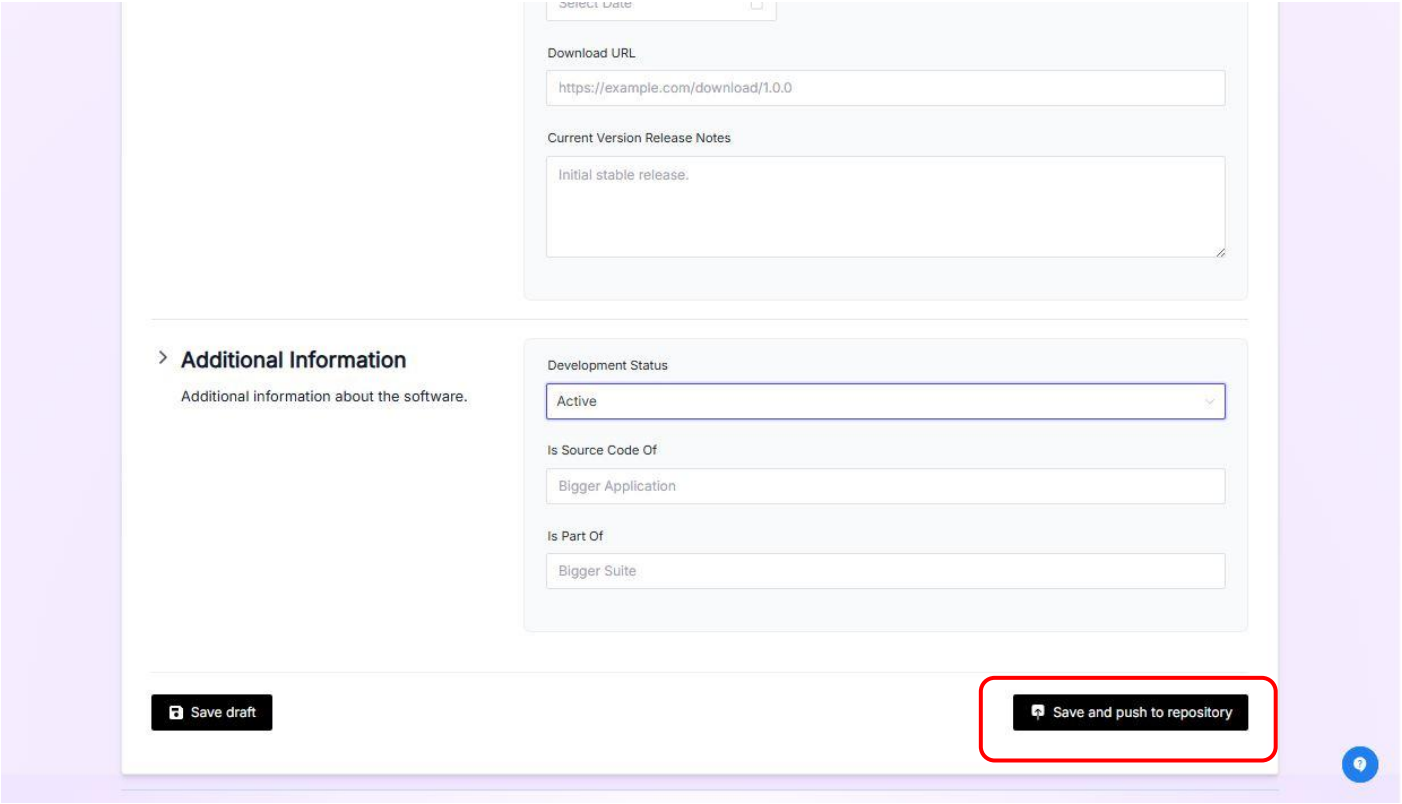
Information about the authors and contributors of the software.

Authors *

Bhavesh Patel

Remove Author

Codefair
Demo



Codefair

Demo

FAIR-BioRS / my-data-analysis-code

Type / to search

Code | Issues 1 | Pull requests 1 | Actions | Projects | Wiki | Security | Insights | Settings

feat: ✨ Add code metadata files #2

Try the new experience | Edit | Code

Open codefair-io wants to merge 2 commits into `main` from `code-metadata-Dt0qs`

Conversation 0 | Commits 2 | Checks 0 | Files changed 2 | +44 −0

Changes from all commits | File filter | Conversations | 0 / 2 files viewed | Review in codespace | Review changes

Filter changed files

- CITATION.cff
- codemeta.json

14 CITATION.cff — Viewed

```
@@ -0,0 +1,14 @@
+ title: my-data-analysis-code
+ authors:
+   - given-names: Bhavesh
+     family-names: Patel
+     affiliation: California Medical Innovations Institute
+ cff-version: 1.2.0
+ message: If you use this software, please cite it using the metadata from this file.
+ type: software
+ abstract: Repo to demo codefair.io
+ keywords:
+   - python
+   - biomedical
+ license: MIT
+ repository-code: https://github.com/FAIR-BioRS/my-data-analysis-code
```

30 codemeta.json — Viewed

```
@@ -0,0 +1,30 @@
+ {
+   "name": "my-data-analysis-code",
+   "@context": "https://w3id.org/codemeta/3.0",
+   "author": [
+     {
+       "affiliation": {
+         "name": "California Medical Innovations Institute",
+         "type": "Organization"
+       },
+       "familyName": "Patel",
+       "id": "_:author_1",
+       "givenName": "Bhavesh",
+       "type": "Person"
+     }
```

Codefair

Demo

FAIR-BioRS / my-data-analysis-code

Type / to search

Code | Issues 1 | Pull requests 1 | Actions | Projects | Wiki | Security | Insights | Settings

FAIR Compliance Dashboard #1

Edit | New issue

Open

codefair-io bot opened 5 minutes ago · edited by codefair-io

Edits | Contributor

Check the FAIRness of your software

This issue is your repository's dashboard for all things FAIR. Keep it open as making and keeping software FAIR is a continuous process that evolves along with the software. You can read the documentation to learn more.

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Edit README

LICENSE ✔

A `LICENSE` file is found at the root level of the repository.

Edit License

Metadata ✔

A `CITATION.cff` and `codemeta.json` file are found in the repository. They may need to be updated over time as new people are contributing to the software, etc.

Edit Metadata

FAIR Software Release ✘

To make your software FAIR, it is necessary to archive it in an archival repository like Zenodo every time you make a release. When you are ready to make your next release, click the "Create release" button below to easily create a FAIR release where your metadata files are updated (including with a DOI) before creating a GitHub release and archiving it.

Create Release

Additional Recommendations

Assignees
No one - Assign yourself

Labels
No labels

Type
No type

Projects
No projects

Milestone
No milestone

Relationships
None yet

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Code with agent mode
Create a branch for this issue or link a pull request.

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Subscribe
You're not receiving notifications from this thread.

Participants
No participants

Transfer issue
Duplicate issue

Codefair

Demo

Codefair

About FAIR Software Dashboard

Dashboard / FAIR-BioRS / my-data-analysis-code / Release / Zenodo

FAIR Software Release

Need help?

Create a release of your repository on Zenodo. We will also create a Github release and update all referencing DOIs to point to the Zenodo release.

Required metadata files

Confirm License

A license file is required for the repository to be released on Zenodo.

Review license

The license file was identified as `MIT`

I have added and reviewed the license file that is required for the repository to be released on Zenodo.

Confirm metadata

A code metadata file is required for the repository to be released on Zenodo.

Review code metadata

Your codemeta.json file is used to generate the title, description, and metadata of your Zenodo deposition. Please make sure that the content of this file is correct and up-to-date.

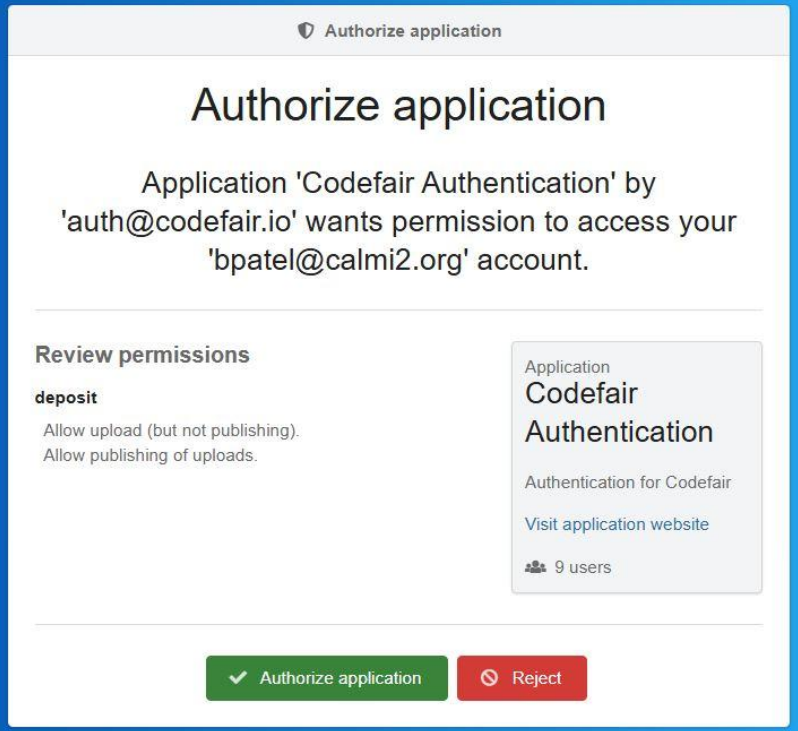
Codefair will automatically add/update the version number, last modified date, and software identifier before the release so you do not need to worry about them.

I have added and reviewed the `citation.CFF` and `codemeta.json` files. I have verified that the content of these files are correct and up-to-date.

You have not yet confirmed the required metadata files. Please review the license and code metadata files above to continue.

Codefair

Demo



Codefair

Demo

Zenodo metadata

Provide repository-specific metadata

Your `codemeta.json` file is used to generate the title, description, and metadata of your Zenodo deposition. Please make sure that the content of this file is correct and up-to-date. Provide below metadata required by Zenodo but not available in codemeta.json.

Version number for this release *

1.0.0

Access Right *

- Open Access
- Embargoed Access
- Restricted Access
- Closed Access

Confirm metadata

Zenodo release

Publish release on Zenodo

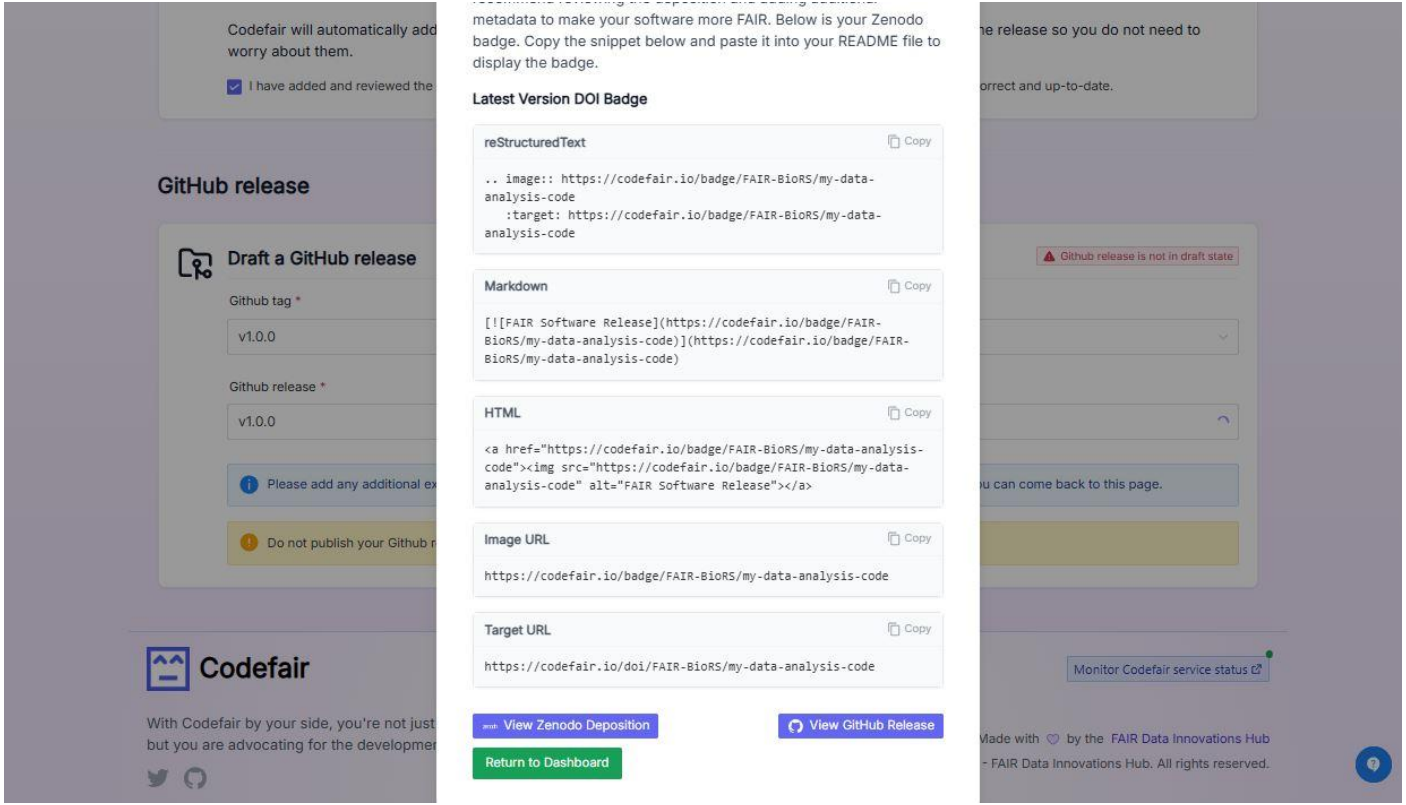
You may save the current configuration as a draft and come back to it later.

Save draft

Start the Zenodo publish process

Codefair

Demo



Codefair

Demo

Open FAIR Compliance Dashboard #1

Edit Metadata

FAIR Software Release

v1.0.0 of your software was successfully released on GitHub and archived on Zenodo. You can view the Zenodo archive by clicking the button below:

DOI 10.5281/zenodo.17238385

Ready to create your next FAIR release? Click the button below:

Create Release

Additional Recommendations

Although these files aren not part of the core FAIR compliance checks, Codefair recommends including them to improve project governance, community engagement, and contributor experience:

CONTRIBUTING.md

This file helps communicate contribution processes and gives contributors clear guidelines placed at the repo root (or in docs/ or .github/), saving time and reducing rework. It surfaces automatically on pull requests, issues, and the repository's Contribute page to guide contributors at every step.

Edit CONTRIBUTING

CODE_OF_CONDUCT.md

Defines clear standards for respectful engagement and shows a welcoming, inclusive community by outlining expectations and procedures for handling abuse. When placed at the repo root (or in docs/ or .github/), CODE_OF_CONDUCT.md surfaces in the repository's community profile and contributor pages to guide behavior at every step.

Edit CODE OF CONDUCT

Last updated Sep 30 2025, 14:11:06 (timezone: America/Los_Angeles)

Create sub-issue

Assignees
No one - Assign yourself

Labels
No labels

Type
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Duplicate issue
Lock conversation
Pin issue
Delete issue

Add a comment

Write Preview

Use Markdown to format your comment

Codefair

Future work

Promote Codefair (currently installed on 760+ repo across 40 users/orgs)

Add more features (README improvement, code formatting, CHANGELOG) and more integrations (GitLab, CI/CD)

Align with the Actionable FAIR4RS guidelines when they are available

Get funding to move this from a side project to an actively developed and sustainable platform

Thank You!

bpatel@calmi2.org

fairdataihub.org

Find these slides and all resources here

bit.ly/codefair-SCW

This work is supported by Azure sponsorship credits granted by Microsoft's AI for Good Research Lab